

CAMPUSCONNECT – REST API ENDPOINT SPECIFICATION

Complete Route Definitions, Payloads, HTTP Status Codes & Access Controls

SLIIT SE2020: Web and Mobile Technologies | Year 2 Semester 2 — 2026

REST API SPECIFICATION & STATUS CODES

Base URL: <https://campusconnect-api.onrender.com/api> (Local: <http://localhost:5000/api>)

All endpoints follow standard response envelope: { success: boolean, message?: string, data?: object }

1. SYSTEM HEALTH & AUTHENTICATION

- **GET /api/health**
 - Access: Public | Returns: 200 OK | Description: Uptime monitoring & server status.
- **POST /api/auth/register**
 - Access: Public | Body: { name, email, password, isAdmin? } | Returns: 201 Created, 400, 409
 - Description: Creates a new student/admin account; hashes password with bcrypt (10 rounds).
- **POST /api/auth/login**
 - Access: Public | Body: { email, password } | Returns: 200 OK, 400, 401
 - Description: Compares password hash, signs JWT token (30d expiry), returns user info.
- **GET /api/auth/me**
 - Access: Protected (Bearer JWT) | Returns: 200 OK, 401 | Description: Current authenticated user.

2. PRIMARY ENTITY: ITEM CRUD

- **GET /api/items**
 - Access: Protected | Query: ?type=Lost|Found&category=... | Returns: 200 OK
 - Description: Retrieves all active items with reporter populated (name, email).
- **GET /api/items/:id**
 - Access: Protected | Returns: 200 OK, 404 | Description: Comprehensive item details by ID.
- **POST /api/items**
 - Access: Protected | Body: multipart/form-data (title, desc, category, location, image)
 - Returns: 201 Created, 400 | Description: Creates item report, stores photo via Multer.
- **PUT /api/items/:id**
 - Access: Protected (Owner or Admin) | Returns: 200 OK, 403, 404 | Description: Updates item details.
- **DELETE /api/items/:id**
 - Access: Protected (Owner or Admin) | Returns: 200 OK, 403, 404
 - Description: Removes item record and cascades deletion to associated claims.

3. RELATED ENTITY: CLAIM CRUD & BUSINESS LOGIC

- **POST /api/claims**
 - Access: Protected (Students) | Body: { itemId, message } | Returns: 201 Created, 400, 409
 - Business Rules: Rejects Lost items (Rule 1), rejects non-Active (Rule 2), prevents duplicates (Rule 3).
- **GET /api/claims**
 - Access: Protected | Query: ?status=Pending | Returns: 200 OK
 - Description: Normal user sees own claims; Administrator sees all submitted campus claims.
- **GET /api/claims/:id**
 - Access: Protected (Owner or Admin) | Returns: 200 OK, 403, 404 | Description: Populated claim view.
- **PUT /api/claims/:id/status**
 - Access: Admin Only | Body: { status: 'Approved' | 'Rejected' } | Returns: 200 OK, 400, 403
 - Rule 4: Approving sets Claim -> Approved AND Item -> Claimed. Locks future claims (Rule 7).
 - Rule 5: Rejecting sets Claim -> Rejected, Item remains Active for other claimants.
- **PUT /api/claims/:id/cancel**
 - Access: Protected (Claimant Only) | Returns: 200 OK, 400, 403
 - Rule 6: Student cancels pending claim; sets status -> Cancelled; Item remains Active.
- **DELETE /api/claims/:id**
 - Access: Protected (Owner or Admin) | Returns: 200 OK, 403, 404 | Description: Deletes claim.